EX NO: 8 SIMULATION AND ANALYSIS OF BPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB

i) To generate BPSK symbols from a binary data sequence using phase-shift keying principles.

ii) To Plot and analyze the BPSK signal constellation in the In-Phase (I) and Quadrature (Q) plane.

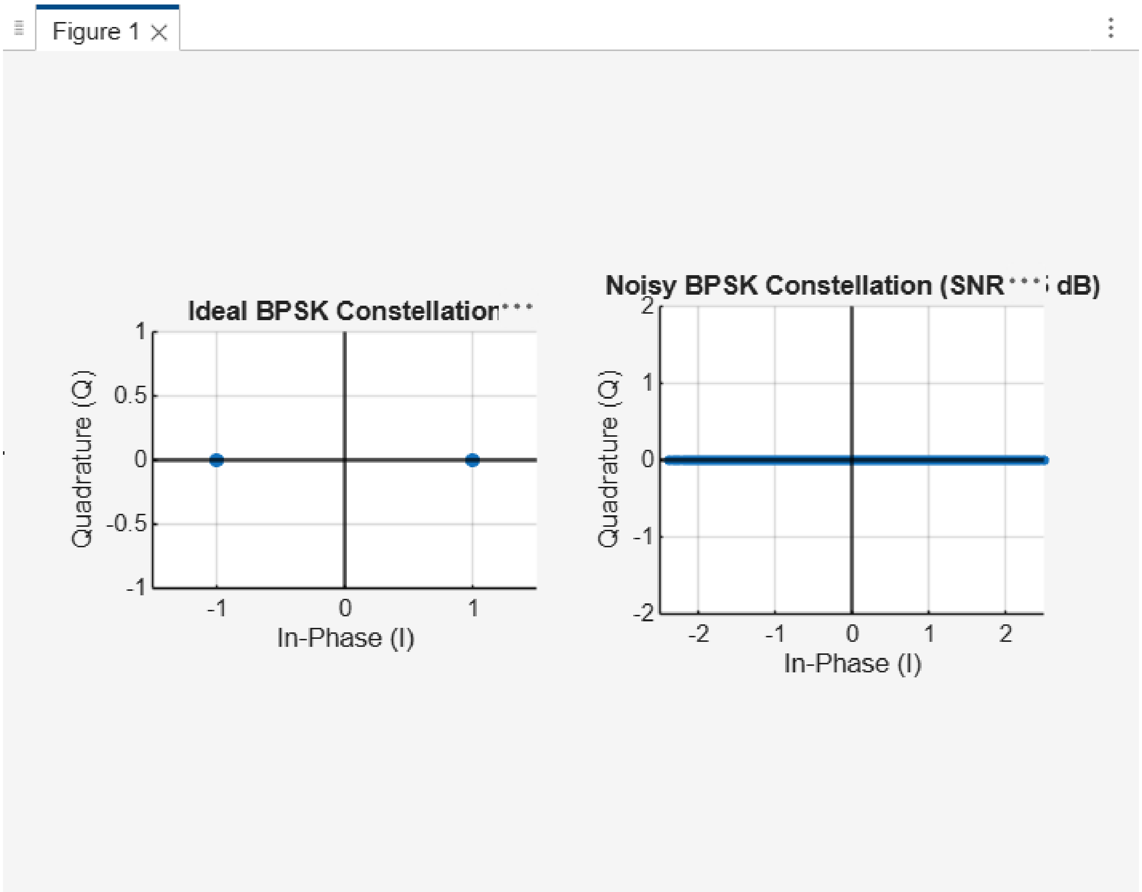
iii) To investigate the effect of noise on constellation points and evaluate the resulting degradation

in symbol detection performance.

```
clc;
clear;
close all;
%% Objective 1: Generate BPSK symbols
N = 2000; % Number of bits
data = randi([0 1], N, 1); % Random binary sequence
% BPSK Mapping
bpsk_symbols = 2*data - 1;
%% Objective 2: Plot BPSK Signal Constellation
figure;
subplot(1,2,1);
scatter(real(bpsk_symbols), imag(bpsk_symbols), ...
    30, 'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlim([-1.5 1.5]);
ylim([-1 1]);
```

```
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal BPSK Constellation');
%% Objective 3: Add Noise and Evaluate Performance
SNR = 5; % Signal-to-Noise Ratio (dB)
rx_signal = awgn(bpsk_symbols, SNR, 'measured');


subplot(1,2,2);
scatter(real(rx_signal), imag(rx_signal), ...
    15, 'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlim([-2.5 2.5]);
ylim([-2 2]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy BPSK Constellation (SNR = ', num2str(SNR), ' dB)']);
%% Symbol Detection
detected_bits = real(rx_signal) > 0;
% Calculate Errors
num_errors = sum(data ~= detected_bits);
BER = num_errors/N;
fprintf('Number of Transmitted Bits : %d\n', N);
fprintf('Number of Bit Errors : %d\n', num_errors);
fprintf('Bit Error Rate (BER) : %f\n', BER);
```



RESULT: The effect of AWGN noise on the constellation points was successfully analyzed, and the resulting degradation in symbol detection performance was evaluated using the Bit Error Rate (BER).

EX NO: 8b SIMULATION AND ANALYSIS OF QPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB

Objective:

i) To generate QPSK symbols from a binary data sequence using quadrature phase-shift keying principles.

ii) To plot and analyze the QPSK signal constellation in the In-Phase (I) and Quadrature (Q) plane.

iii) To investigate the effect of noise on constellation points and evaluate the resulting degradation in symbol detection performance.

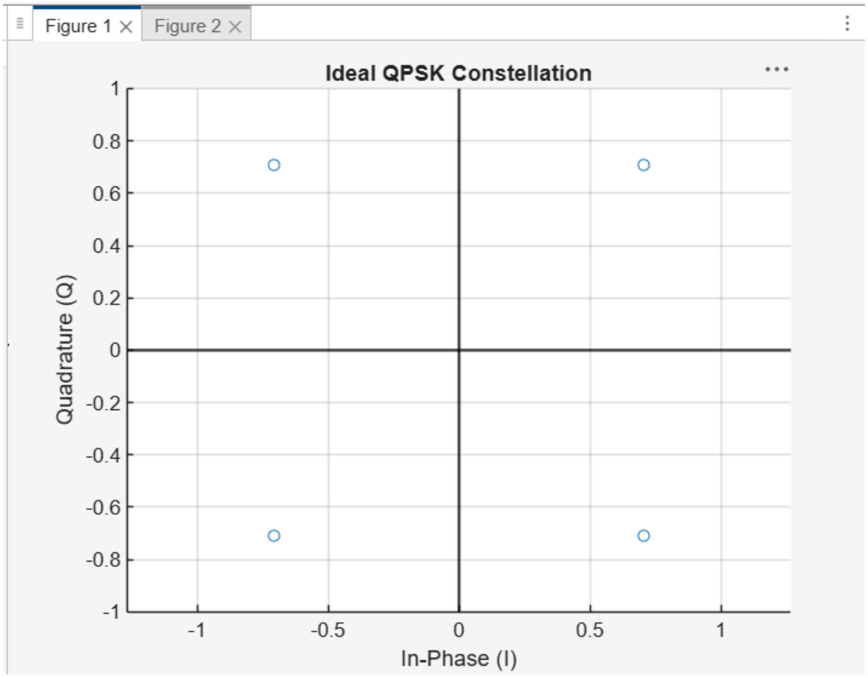
CODE:

```
clc;
clear;
close all;
%% Objective 1: Generate QPSK Symbols
N = 1000; % Number of bits
data = randi([0 1], N, 1); % Random binary sequence
% Ensure even number of bits
if mod(N,2)~=0
data = [data; 0];
N = N + 1;
end
% Group bits into pairs
bit_pairs = reshape(data,2,[])';
% QPSK Mapping
qpsk_symbols = zeros(size(bit_pairs,1),1);
for k = 1:size(bit_pairs,1)
if isequal(bit_pairs(k,:),[0 0])
qpsk_symbols(k) = (1+1j)/sqrt(2);
elseif isequal(bit_pairs(k,:),[0 1])
qpsk_symbols(k) = (-1+1j)/sqrt(2);
elseif isequal(bit_pairs(k,:),[1 1])
qpsk_symbols(k) = (-1-1j)/sqrt(2);
elseif isequal(bit_pairs(k,:),[1 0])
```

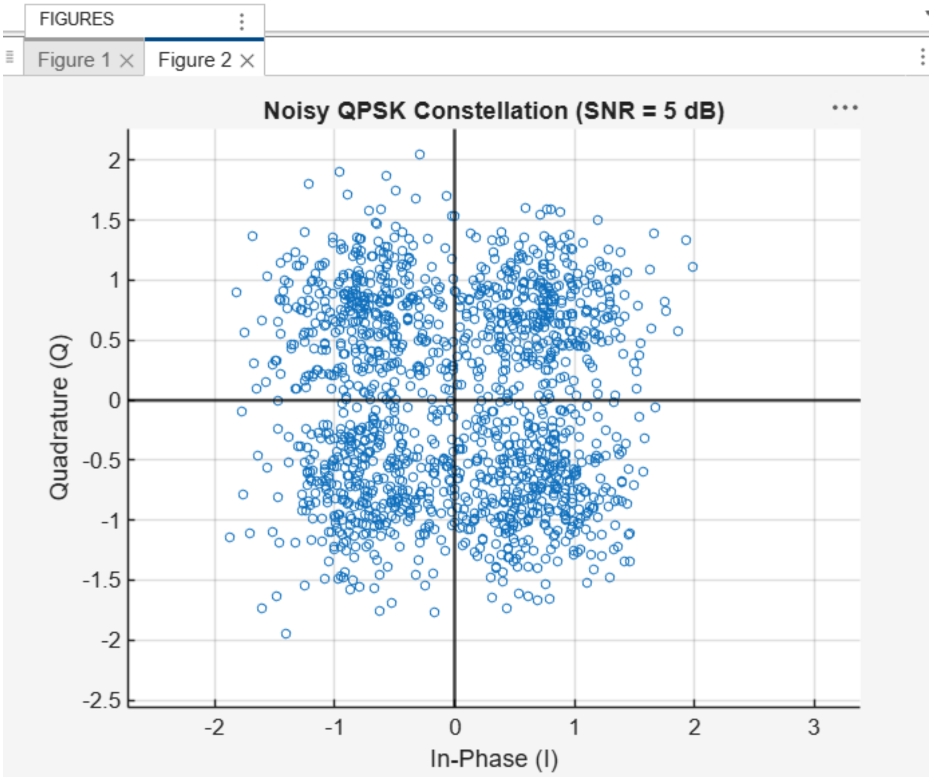
```
qpsk_symbols(k) = (1-1j)/sqrt(2);
end
end
%% Objective 2: Plot QPSK Constellation
scatter(real(qpsk_symbols),imag(qpsk_symbols), 30);
xlim([-1 1])
ylim([-1 1])
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal QPSK Constellation');
%% Objective 3: Add Noise and Evaluate Performance
SNR = 5;
rx_signal = awgn(qpsk_symbols,SNR,'measured');
figure;
scatter(real(rx_signal),imag(rx_signal), 15);
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy QPSK Constellation (SNR = ',num2str(SNR),' dB)']);
%% Symbol Detection
detected_bits = zeros(length(rx_signal)*2,1);
for k = 1:length(rx_signal)
I = real(rx_signal(k));
```

```
Q = imag(rx_signal(k));
if I>0 && Q>0
bits = [0 0];
elseif I<0 && Q>0
bits = [0 1];
elseif I<0 && Q<0
bits = [1 1];
else
bits = [1 0];
end
detected_bits(2*k-1:2*k) = bits;
end
%% BER Calculation
num_errors = sum(data ~= detected_bits);
BER = num_errors/N;
fprintf('Number of Transmitted Bits : %d\n',N);
fprintf('Number of Bit Errors : %d\n',num_errors);
fprintf('Bit Error Rate (BER) : %f\n',BER);
```

ob1



Ob2



Number of Transmitted Bits : 3000
Number of Bit Errors : 97
Bit Error Rate (BER) : 0.032333

>>

Result:

The effect of AWGN noise on the QPSK constellation points was successfully analyzed, and the resulting degradation in symbol detection performance was evaluated using the Bit Error Rate (BER).